OBESITY

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2025-2026

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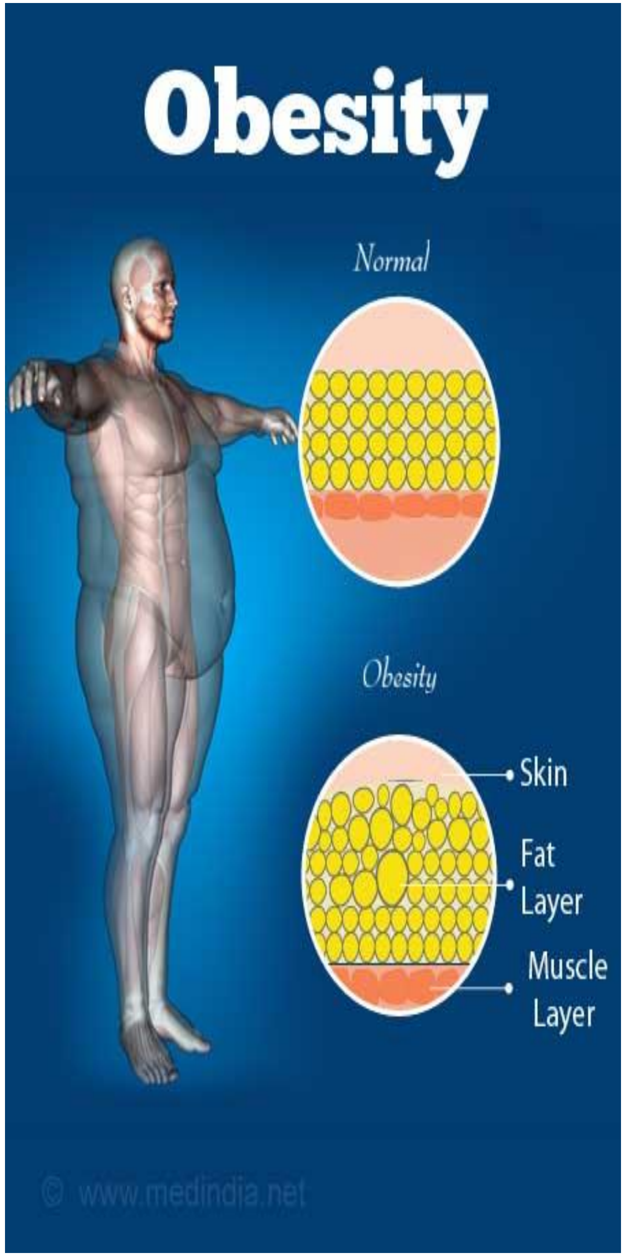
Learning Outcomes:

- Define Obesity.
- Calculate the BMI.
- Outline the methods for diagnosis obesity.
- Identify the contributing factors of Obesity.
- Explain the physiological mechanism of food intake regulation.
- Describe some treatment modalities of obesity.

- When the energy content of food balances the energy used by all the cells of the body, body weight remains constant (unless there is a gain or loss of water).

- For each 9.3 calories of excess energy that remains in the body, approximately 1 gram of fat is stored.



Diagnosis of Obesity

- Obesity is defined as 25% or greater total body fat in men and 35% or greater total body fat in women.
- A surrogate marker for body fat content is the body mass index (BMI), which is calculated as: BMI Weight in kilograms divided by Height in meters = 2.
- In clinical terms, a person with a BMI between 25 and 29.9 kg/m2 is deemed overweight, and a person with a BMI greater than or equal to 30 kg/m2 is considered obese.

$$\text{BMI} = \frac{\text{weight (kg)}}{\text{height (m}^2\text{)}}$$

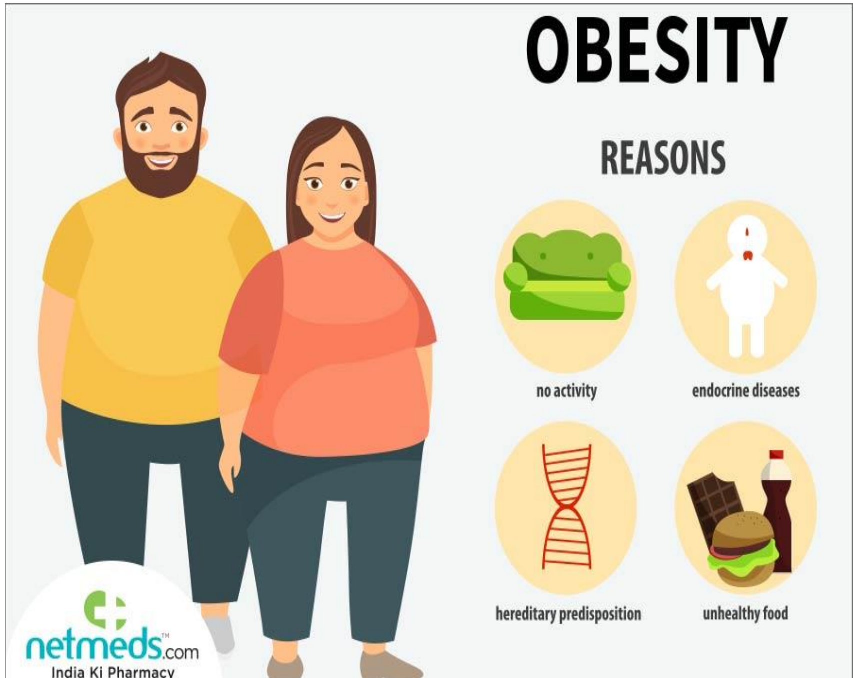
Diagnosis of Obesity

- BMI is not a direct estimate of adiposity, individuals have a high BMI as a result of a large muscle mass.

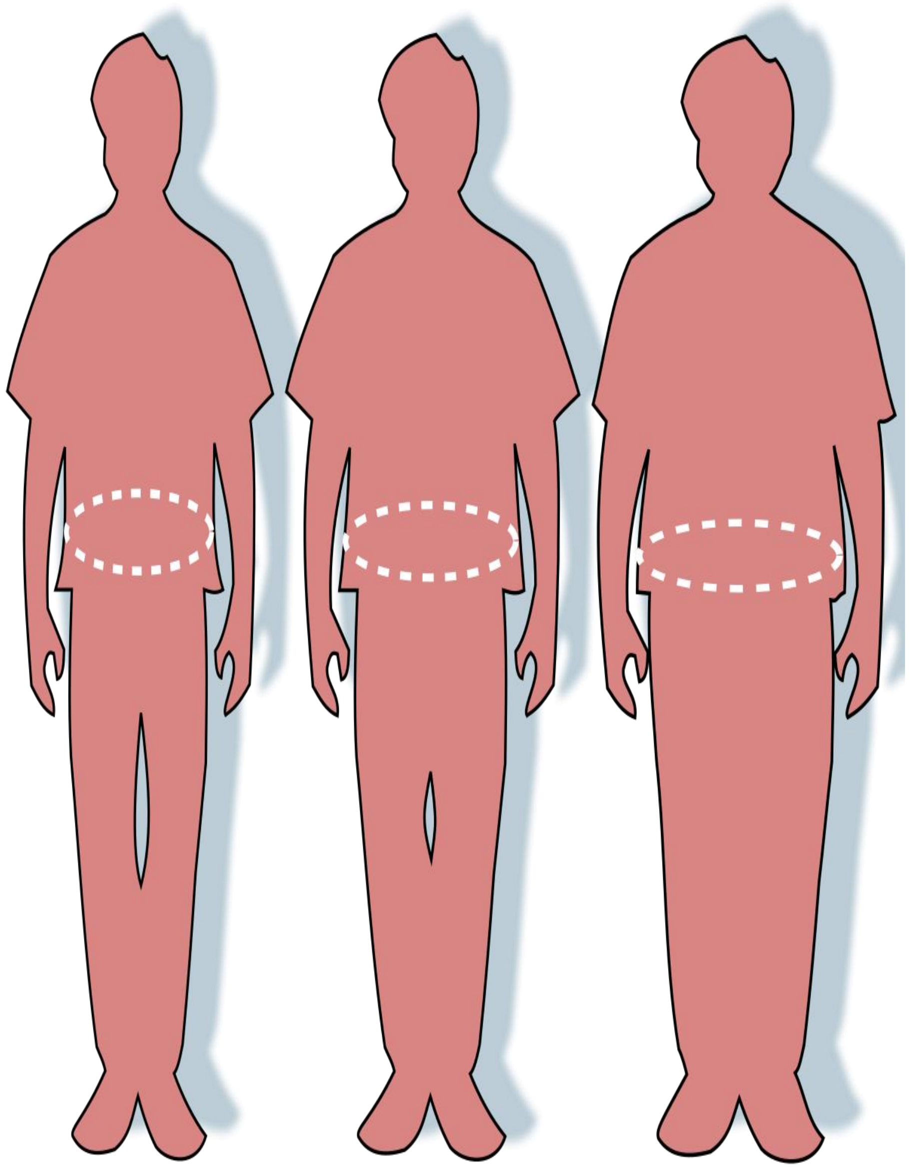
- A better way to define obesity is to actually measure the percentage of total body fat.

- Percentage of body fat can be estimated with various methods bioelectrical impedance.

- Being overweight and Even moderate obesity is hazardous to health; it is a risk factor in cardiovascular disease, hypertension, diabètes mellites, arthritis and certain cancers.



- Initially, the adipocytes increase in size, but at a maximal size, they divide and proliferation of adipocytes occurs in extreme obesity.
- The enzyme endothelial lipoprotein lipase regulates triglyceride storage.
- The enzyme is very active in abdominal fat but less active in hip fat.
- Accumulation of fat in the abdomen is associated with higher blood cholesterol level and other cardiac risk factors because adipose cells in this area appear to be more metabolically active.



Contributing factors of Obesity:

- **Contributing factors of Obesity** include:

- ❑ Genetic factors: *mutations of POMC and MCR-4* genes.
- ❑ Eating habits taught early in life.
- ❑ overeating to relieve tension.
- ❑ Social customs.
- ❑ **Sedentary Lifestyle Is a Major Cause of Obesity.**

- In a few cases, obesity may result from:

trauma of or tumors in the food-regulating centers in the hypothalamus. In many cases of obesity, no specific cause can be identified.

Neural connections between the hypothalamus and other parts of the brain and their role in Food Intake Regulation:

- Within the hypothalamus are clusters of neurons that play key roles in regulating food intake.
- **Satiety** is a feeling of fullness accompanied by lack of desire to eat.
- Two hypothalamic areas involved in regulation of food intake are the *arcuate nucleus* and the *paraventricular nucleus* .
- There is a mouse gene, named *obese*, that causes overeating and severe obesity in its mutated form.
- The product of this gene is the hormone **Leptin**.
- Leptin helps decrease **adiposity (**total body-fat mass).

- Leptin is synthesized and secreted by adipocytes in proportion to adiposity; as more triglycerides are stored, more leptin is secreted into the bloodstream.

- Leptin acts on the hypothalamus to inhibit circuits that stimulate eating while also activating circuits that increase energy expenditure.

- The hormone insulin has a similar, but smaller, effect. Both leptin and insulin are able to pass through the blood–brain barrier.

- When leptin and insulin levels are *low*, neurons that extend from the arcuate nucleus to the paraventricular nucleus release a neurotransmitter called **neuropeptide Y** that stimulates food intake.

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- Other neurons that extend between the arcuate and paraventricular nuclei release a neurotransmitter called **melanocortin.**

- Leptin stimulates release of melanocortin.

- **melanocortin** acts to inhibit food intake.

- Although leptin, neuropeptide Y, and melanocortin are key signaling molecules for maintaining energy homeostasis, several other hormones and neurotransmitters also contribute.

Feedback Mechanism In Regulation of Food Intak:

- Many experiments have demonstrated the presence of satiety signals, chemical or neural changes that help terminate eating when “fullness” is attained:

1. an increase in blood glucose level, as occurs after a meal, decreases appetite
2. Several hormones, such as glucagon.
3. Distension of the GI tract, particularly the stomach and duodenum, also contributes to termination of food intake.

Treatment of Obesity:

1- Behavior modification programs:

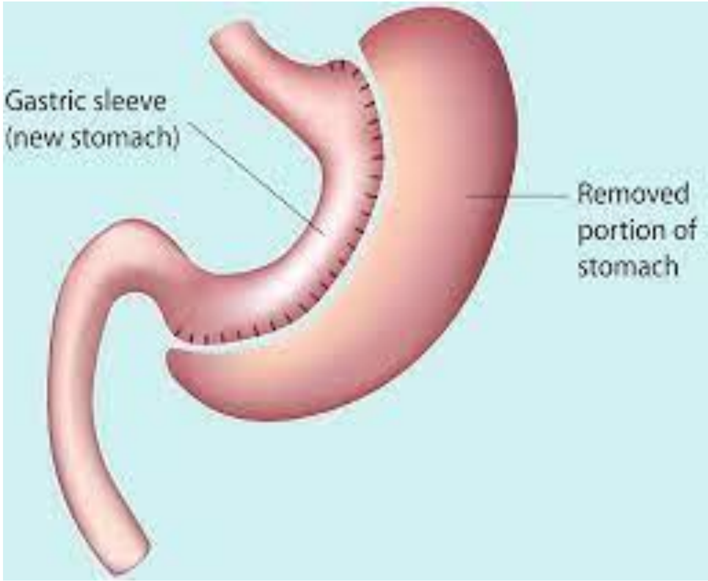
- Alter eating behaviors and increase exercise activity.
- Even a single episode of strenuous exercise may increase basal energy expenditure for several hours after the physical activity is stopped.

2- Very-low-calorie diets.

Treatment of Obesity:

3- Drugs e.g. Orlistat works by inhibiting the lipases released into the lumen of the GI tract. With less lipase activity, fewer dietary triglycerides are absorbed.

4- Surgery: a surgical procedure may be considered. Ex. *vertical sleeve gastrectomy* which reduce the stomach size so that it can hold just a tiny quantity of food.



THANK YOU!!!